NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

01 July 2026

161 verified · 260 raw · 6 sources

161	3	158	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	138	Media
TechCrunch AI	11	Media
HackerNews	8	Community
The Verge AI	4	Media
MIT Tech Review	3	Media
VentureBeat AI	1	Media

VERIFIED INTELLIGENCE — 3 stories

VERI

Anthropic’s long-sidelined Fable 5 is greenlit to return

The Verge AI +2 sources 100%

After weeks of negotiating with the Trump administration, Anthropic is finally going to be able to bring Claude Fable 5 back online. In a post on X, Anthropic said it plans to begin restoring access Wednesday to users globally on Claude platforms, and that the company would...

https://www.theverge.com/ai-artificial-intelligence/958964/anthropic-claude-fable-5-is-...

VERI

Show HN: Fleet – a local-first console for managing Dockerized Hermes AI Agents

HackerNews +1 source 90% HN 8

arXiv:2606.31693v1 Announce Type: cross Abstract: The wave of AI-native applications is moving shopping beyond page- and feed-based browsing toward intent-driven experiences orchestrated by LLM agents. A common design wraps an LLM around existing search and recommendation...

https://github.com/matt454/agent-fleet-console

VERI

Histogram-constrained Image Generation

ArXiv CS.AI +1 source 80%

arXiv:2606.31683v1 Announce Type: cross Abstract: Diffusion models have emerged as a dominant paradigm in generative modeling, enabling high-fidelity sampling from complex data distributions. Despite impressive capabilities, controlling diffusion models to produce outputs...

https://arxiv.org/abs/2606.31683

CONFIRMED SIGNALS — 158 stories

CONF

Show HN: I made a heatmap of 3400 VCs who are open to cold emails

HackerNews 50% HN 23

No summary — see link for full article.

https://apparent.social/heat-map

CONF

Show HN: TraceAIO – open-source LLM visibility tracker

HackerNews 50% HN 6

I built TraceAIO, an open-source tool that prompts LLMs on your behalf and tells you whether ChatGPT, Perplexity, and Gemini mention your brand — and which competitors and sources show up instead. Yeah, this category smells a bit like a grift, same as early SEO. And I think over...

https://traceaio.org

CONF

Show HN: Privacy policy generator for AI apps (LLM disclosure, EU AI Act)

HackerNews 50% HN 6

No summary — see link for full article.

https://ai-policy-gen.pages.dev

CONF

Show HN: Makes local LLMs faster and more reliable by optimizing for your device

HackerNews 50% HN 5

Time to first token is 39% faster Agent wall times decrease by 46% No swaps Tracks your resource usage in real-time and adjusts how the model runs so that it works perfectly on your device. Implements KV cache sizing, prefix caching, live RAM pressure management, context...

https://www.autotunellm.com/

CONF

Show HN: AMA2, messenger built for AI agent

HackerNews 50% HN 5

I'm a solo founder building AMA2, a messaging runtime made for AI agents. This is my first Show HN, so I'd really appreciate your feedback. What brought me this idea: At first, I was building an AI agent for solo creators that knows everything about you and can do business chore...

https://ama2.me/

CONF

Show HN: Don't ask if devs cheat with AI, test if they're good with it

HackerNews 50% HN 5

No summary — see link for full article.

https://tryevaluator.com

0 CONF

Show HN: fenic – LLMs as dataframe operators, query meaning and structure

HackerNews 50% HN 3

Hey friends. I'd like to share a project that's dear to me. fenic is a dataframe API with LLMs added as first-class citizens, a classic lazy dataframe API extended with new operators that are backed by LLMs. What this gets you is the ability to work with structured and...

https://github.com/typedef-ai/fenic

1 CONF

Trump drops restrictions on Anthropic's Mythos and Fable models

TechCrunch AI 50%

The Trump administration's erratic approach to AI policymaking has left companies across the industry with little clarity about what will govern future model releases.

https://techcrunch.com/2026/06/30/trump-drops-restrictions-on-anthropics-mythos-and-fab...

2 CONF

OpenClaw is finally available on Android and iOS

TechCrunch AI 50%

The free open source agentic program is finally invading your phone.

https://techcrunch.com/2026/06/30/openclaw-is-finally-available-on-android-and-ios/

3 CONF

The DeepMind trio who built a poker AI are now making money for quant hedge funds

TechCrunch AI 50%

EquiLibre Technologies, a Prague-based AI lab founded by three ex-DeepMind researchers, is now valued at more than $500 million.

https://techcrunch.com/2026/06/30/the-deepmind-trio-who-built-a-poker-ai-are-now-making...

4 CONF

Nvidia competitor Etched hits $5B valuation, $1B in sales for AI chip

TechCrunch AI 50%

Nvidia AI chip competitor Etched says it has already booked $1 billion under contract for the inference systems powered by its chip.

https://techcrunch.com/2026/06/30/nvidia-competitor-etched-hits-5b-valuation-1b-in-sale...

5 **CONF**

X now offers an MCP server to make its platform easier for AI tools to use

TechCrunch AI **50%**

X has launched a hosted MCP server, making it easier for developers to connect AI applications with the company’s API.

https://techcrunch.com/2026/06/30/x-now-offers-an-mcp-server-to-make-its-platform-easie...

6 **CONF**

Amazon launches new $1 billion FDE org, following OpenAI and Anthropic

TechCrunch AI **50%**

Engineers on the new team will embed within companies to deploy purpose-built agents, focusing on fast deployments and customer self-sufficiency.

https://techcrunch.com/2026/06/30/amazon-launches-new-1-billion-fde-org-following-opena...

7 **CONF**

Lumo, Proton’s privacy-focused AI chatbot, gets an upgrade

TechCrunch AI **50%**

Proton's Lumo 2.0 is dropping this week, giving users a broader variety of capabilities.

https://techcrunch.com/2026/06/30/lumo-protons-privacy-focused-ai-chatbot-gets-an-upgrade/

8 **CONF**

Crypto exchange OKX wants AI agents to hire and pay each other

TechCrunch AI **50%**

OKX is bringing together payments, identity, and reputation into a marketplace for AI agents.

https://techcrunch.com/2026/06/30/crypto-exchange-okx-wants-ai-agents-to-hire-and-pay-e...

9 **CONF**

Vibe-coding platform Base44 launches own model as AI startups seek defensibility

TechCrunch AI **50%**

Wix-owned vibe-coding platform Base44 has started rolling out its own AI model — with hopes that it will eventually outperform frontier models.

https://techcrunch.com/2026/06/29/vibe-coding-platform-base44-launches-own-model-as-ai-...

0 **CONF**

Claude Science is Anthropic’s newest flagship product

MIT Tech Review **50%**

At an event for pharmaceutical executives, biotech founders, and researchers on Tuesday, Anthropic announced Claude Science, a major new product intended to support scientific research in the same way that Claude Code supports software engineering. Like Claude Code, Claude...

https://www.technologyreview.com/2026/06/30/1139987/claude-science-is-anthropics-newest...

1 **CONF**

The Download: AI “coworkers” and stratospheric internet

MIT Tech Review **50%**

This is today’s edition of The Download, our weekday newsletter that provides a daily dose of what’s going on in the world of technology. AI agents are not your “coworkers” Imagine coming in to work to learn that a new underling will report to you. The worker is not a person but...

https://www.technologyreview.com/2026/06/30/1139954/the-download-ai-agents-coworkers-so...

2 **CONF**

Agriculture is ready for AI, but its data isn’t

MIT Tech Review **50%**

Artificial intelligence is transforming what is possible in agriculture, but industry leaders should be wary of investing in AI without first laying the groundwork. The use cases are promising, especially for an industry navigating volatile fertilizer costs, unpredictable...

https://www.technologyreview.com/2026/06/30/1139513/agriculture-is-ready-for-ai-but-its...

3 CONF

Netflix is using an AI-generated Gene Wilder voice in its Willy Wonka reality show

The Verge AI 50%

A new teaser trailer confirmed that Wonka's The Golden Ticket will premiere on Netflix on September 23rd, following its Squid Game reality show in the trend of creating real competitions based on fictional torture scenarios. While the sets seen in the trailer are real and not...

https://www.theverge.com/streaming/959684/netflix-wonka-golden-ticket-gene-wilder

4 CONF

Libby will filter out AI content, kind of

The Verge AI 50%

This is Lowpass by Janko Roettgers, a newsletter on the ever-evolving intersection of tech and entertainment, syndicated just for The Verge subscribers once a week. "AI is the new frontier for us," says Marc DeBevoise, who took over as the new CEO of OverDrive last week....

https://www.theverge.com/column/959433/libby-ai-filter

5 CONF

Tidal won’t pay royalties on AI-generated music, but isn’t banning it outright

The Verge AI 50%

Tidal shared its new policies regarding AI-generated music today and how the platform plans to "protect artists" and "inform listeners." Instead of banning it outright, starting on July 15th Tidal will label tracks it has identified as being 100 percent AI-generated with an...

https://www.theverge.com/tech/959211/tidal-ai-music-policy-demonetizingdetect-label

6 CONF

How Can AI Find My Model? A Model-Finding Experimental Study Considering Data Formats, Embeddings, and Retrieval Strategies

ArXiv CS.AI 50%

arXiv:2606.30846v1 Announce Type: new Abstract: Discovering simulation models for reuse remains a fundamental challenge in Modeling and Simulation (M&S). When many models coexist, identifying those that align with a given modeling intent remains difficult. Recent advances in...

https://arxiv.org/abs/2606.30846

7 CONF

BayesBench: Evaluating LLM Belief Trajectories Under Multi-Turn Evidence Accumulation

ArXiv CS.AI 50%

arXiv:2606.30850v1 Announce Type: new Abstract: Large language models (LLMs) are typically deployed in multi-turn conversations, where each turn provides new evidence that should reduce epistemic uncertainty about their environment. Acting rationally then requires inferring the...

https://arxiv.org/abs/2606.30850

8 CONF

RoPoLL: Robust Panel of LLM Judges

ArXiv CS.AI 50%

arXiv:2606.30931v1 Announce Type: new Abstract: The LLM Jury, a Panel of LLM Evaluators (PoLL) reporting consensus scores, has become a practical alternative to single-judge LLM evaluation, yet its statistical behavior remains poorly understood. We formalize the LLM Jury under...

https://arxiv.org/abs/2606.30931

9 **CONF**

Neuro-Bayesian-Symbolic Residual Attention Shallow Network: Explainable Deep Learning for Cybersecurity Risk Assessment

ArXiv CS.AI **50%**

arXiv:2606.30953v1 Announce Type: new Abstract: We introduce the Neuro-Bayesian-Symbolic Residual Attention Shallow Network (NBS-RASN), a hybrid neural architecture for explainable cybersecurity risk assessment in open-source ecosystems. Unlike deep models that trade...

https://arxiv.org/abs/2606.30953

0 **CONF**

HyPOLE: Hyperproperty-Guided Multi-Agent Reinforcement Learning under Partial Observation

ArXiv CS.AI **50%**

arXiv:2606.30966v1 Announce Type: new Abstract: Formal specification is a powerful tool to guide the learning process and provides significant advantages over reward shaping: (1) mathematical rigor; (2) expressiveness to specify objectives and constraints, and (3) the ability to...

https://arxiv.org/abs/2606.30966

1 **CONF**

OpenLife: Toward Open-World Artificial Life with Autonomous LLM Agents

ArXiv CS.AI **50%**

arXiv:2606.31046v1 Announce Type: new Abstract: Artificial life has explored life-like behavior on many computational substrates, but mostly in researcher-designed closed worlds. We argue that large language model (LLM) agents, with persistent memory, tool use, network access,...

https://arxiv.org/abs/2606.31046

2 **CONF**

An AI-Based Solution for Secure Service Provisioning in IoT

ArXiv CS.AI **50%**

arXiv:2606.30701v1 Announce Type: cross Abstract: As the Internet of Things (IoT) continues its rapid expansion, the attack surface grows accordingly, with emerging threats targeting smart objects and their interactions. In this evolving landscape, securing service provisioning...

https://arxiv.org/abs/2606.30701

3 **CONF**

Beyond Compilation: Evaluating Faithful Natural-Language-to-Lean Statement Formalization

ArXiv CS.AI **50%**

arXiv:2606.31002v1 Announce Type: new Abstract: Theorem-proving benchmarks evaluate proof search against fixed formal statements, but natural-language-to-Lean formalization must generate the formal statement itself. In this setting, compilation is only a validity check: a Lean...

https://arxiv.org/abs/2606.31002

4 **CONF**

MultiUAV-Plat: An LLM-Oriented Platform, Benchmark and Framework for Multi-UAV Collaborative Task Planning

ArXiv CS.AI **50%**

arXiv:2606.31073v1 Announce Type: new Abstract: Large language models (LLMs) provide a promising interface for high-level robotic task planning, but their use in multi-UAV collaboration remains difficult to evaluate systematically. Existing UAV simulators mainly emphasize...

https://arxiv.org/abs/2606.31073

5 CONF

Revealing Safety-Critical Scenarios for UTM via Transformer

ArXiv CS.AI 50%

arXiv:2606.31114v1 Announce Type: new Abstract: Unmanned Traffic Management (UTM) systems are cloud-based platforms designed to manage and coordinate multiple aerial vehicles remotely. UTM systems are safety-critical which cannot tolerate failures like crash or collision. To...

https://arxiv.org/abs/2606.31114

6 CONF

The Past Is Prologue: A Plug-in Controller for Selective Updates in Sequentially Evolving LLM Memory

ArXiv CS.AI 50%

arXiv:2606.31121v1 Announce Type: new Abstract: Sequentially evolving LLM memory enables agents to reuse past experience, but existing systems usually deploy each locally generated memory update without checking whether it improves future behavior. As a result, updates that help...

https://arxiv.org/abs/2606.31121

7 CONF

Scenario Generation for Testing of Autonomous Driving Systems Using Real-World Failure Records

ArXiv CS.AI 50%

arXiv:2606.31131v1 Announce Type: new Abstract: To ensure safe on-road behavior, pre-deployment testing and failure discovery of Autonomous Driving Systems (ADS) is crucial. Present day simulation based testing methods focus largely on mathematical models for efficient search of...

https://arxiv.org/abs/2606.31131

8 CONF

Cross-Domain Feature Expansion for Tabular Medical Data via Knowledge Graphs Injection

ArXiv CS.AI 50%

arXiv:2606.31171v1 Announce Type: new Abstract: Acquiring comprehensive cross-domain biomedical profiles is often costly and time-consuming, resulting in severe data scarcity in medical research. To address this challenge, we propose MedKGTab, a knowledge-injected framework...

https://arxiv.org/abs/2606.31171

9 CONF

ClawArena-Team: Benchmarking Subagent Orchestration and Dynamic Workflows in Language-Model Agents

ArXiv CS.AI 50%

arXiv:2606.31174v1 Announce Type: new Abstract: Production large language-model (LLM) agents are increasingly deployed not as lone problem-solvers but as managers: a main model creates specialized subagents, delegates work, and orchestrates their parallel, asynchronous returns...

https://arxiv.org/abs/2606.31174

0 CONF

HealthAgentBench: A Unified Benchmark Suite of Realistic Agentic Healthcare Environments for Challenging Frontier AI Agents

ArXiv CS.AI 50%

arXiv:2606.31179v1 Announce Type: new Abstract: As AI agents become increasingly capable of complex, long-horizon reasoning, rigorous and holistic evaluation is essential for measuring progress toward real-world healthcare applications. We introduce HealthAgentBench, a suite of...

https://arxiv.org/abs/2606.31179

1 CONF

AI-Assisted Discovery of Convex Relaxations via Dual Agents

ArXiv CS.AI 50%

arXiv:2606.31182v1 Announce Type: new Abstract: Recent work shows that LLM agents can improve sharp-constant inequalities by searching for extremal constructions, which yield upper bounds. We address the complementary side: a lower bound holds for every admissible function and...

https://arxiv.org/abs/2606.31182

2 CONF

Agentic RAG-VLM: Affordance-Aware Retrieval-Augmented Generation with Self-Reflective Planning for Robotic Grasping

ArXiv CS.AI 50%

arXiv:2606.31200v1 Announce Type: new Abstract: Generalizable robotic grasping in cluttered environments is essential for deploying manipulators in unstructured human spaces, yet existing VLM-based methods rely on visual similarity for object matching, neglecting physical...

https://arxiv.org/abs/2606.31200

3 CONF

Embodied CAD: Solver-Grounded LLM Agents for Parametric B-Rep Assembly Modeling

ArXiv CS.AI 50%

arXiv:2606.31252v1 Announce Type: new Abstract: Large language models can write plausible CAD scripts, but reliable industrial CAD modeling requires more than syntactically valid code: every feature, placement, and assembly relation must be accepted by an exact geometric kernel...

https://arxiv.org/abs/2606.31252

4 CONF

Optimization Algorithms for Joint OFDM Waveform Design and RIS Configuration in 6G Networks: From Convex Relaxation to Foundation Models

ArXiv CS.AI 50%

arXiv:2606.31334v1 Announce Type: new Abstract: Joint OFDM-RIS optimization for 6G is a mixed-integer nonlinear programming (MINLP) problem covering sum-rate maximization, energy efficiency, max-min fairness, and peak-to-average power ratio (PAPR)-constrained objectives....

https://arxiv.org/abs/2606.31334

5 CONF

Automating Cause-Effect Specification with Knowledge Graphs and Large Language Models

ArXiv CS.AI 50%

arXiv:2606.31614v1 Announce Type: cross Abstract: Engineering specifications such as interlocks, alarm rationalization tables, and cause-and-effect (C&E) matrices remain central to process control and safety, yet their creation is still predominantly manual, document-driven, and...

https://arxiv.org/abs/2606.31614

6 CONF

Arena-T2I Hard: Benchmarking and Improving Faithfulness with Dependency-Aware Checklist

ArXiv CS.AI 50%

arXiv:2606.31711v1 Announce Type: new Abstract: Faithfulness -- how precisely a generated image aligns with its prompt -- is increasingly central to the real-world utility of text-to-image (T2I) models. Existing faithfulness benchmarks, however, rely on simple atomic...

https://arxiv.org/abs/2606.31711

7 **CONF**

RAISE: LLM-based Automated Heuristic Design with Robust Adversary Instance Search

ArXiv CS.AI **50%**

arXiv:2606.31801v1 Announce Type: new Abstract: Automated Heuristic Design (AHD) with Large Language Models (LLMs) has shown remarkable progress in discovering high-quality heuristics. However, existing LLM-based AHD methods optimize heuristics for a fixed training instance set...

https://arxiv.org/abs/2606.31801

8 **CONF**

Large Databases Need Small, Open-Weight Language Models

ArXiv CS.AI **50%**

arXiv:2606.31808v1 Announce Type: new Abstract: Language model systems built around proprietary APIs often operate on a token-based cost model. This becomes prohibitively expensive in the context of large databases, where LM-enhanced relational operators can incur costs...

https://arxiv.org/abs/2606.31808

9 **CONF**

An Agentic AI Framework to Accelerate Scientific Discovery in Plant Phenotyping

ArXiv CS.AI **50%**

arXiv:2606.31831v1 Announce Type: new Abstract: High-throughput plant phenotyping now generates image derived datasets far faster than scientists can analyze them. At Oak Ridge National Laboratory's Advanced Plant Phenotyping Laboratory (APPL), automated stations image hundreds...

https://arxiv.org/abs/2606.31831

0 **CONF**

Harnessing Textual Refusal Directions for Multimodal Safety

ArXiv CS.AI **50%**

arXiv:2606.31876v1 Announce Type: new Abstract: To improve safety in Large Language Models (LLMs) we can either perform post-training alignment or exploit refusal directions in the activation space. Both strategies are less feasible in Multimodal LLMs (MLLMs) as they require...

https://arxiv.org/abs/2606.31876

1 **CONF**

TreeAgent: A Generalizable Multi-Agent Framework for Automated Bias Labeling in Forestry via Compiled Expert Rules and Vision-Language Models

ArXiv CS.AI **50%**

arXiv:2606.31976v1 Announce Type: new Abstract: Human-labeled data are widely used as reference annotations in ML, despite known variability across annotators in many expert-driven domains. In addition, expert annotation is slow, inconsistent, and remains a major bottleneck for...

https://arxiv.org/abs/2606.31976

2 **CONF**

Self-Study Reconsidered: The Hidden Fragility of Learning from Self-Generated QA

ArXiv CS.AI **50%**

arXiv:2606.32002v1 Announce Type: new Abstract: Language models are increasingly taught from synthetic question--answer (QA) supervision: a model generates questions about a document, answers them from the same text, and the resulting pairs are used to fine-tune, distill, or...

https://arxiv.org/abs/2606.32002

3 **CONF**

Surrogate-Gated Generation and Foundation-Model Embeddings for Bayesian Materials Design

ArXiv CS.AI 50%

arXiv:2606.28578v1 Announce Type: cross Abstract: Closed-loop materials discovery iterates between proposing candidate structures and evaluating their properties, and property evaluation dominates the cost. In the generative variant, a learned prior proposes candidate crystals...

https://arxiv.org/abs/2606.28578

4 **CONF**

ASR-Agnostic Multimodal Spectrotemporal Modeling for Early Dementia Detection

ArXiv CS.AI 50%

arXiv:2606.30646v1 Announce Type: cross Abstract: Speech recruits the same executive, attentional, and working memory processes underlying instrumental activities of daily living, or IADLs, providing a non-invasive proxy for cognitive assessment. Yet most speech-based dementia...

https://arxiv.org/abs/2606.30646

5 **CONF**

Qualified Educational Capacity Planning under Heterogeneous Student Support Needs: A Synthetic Benchmark and Decision-Support Framework

ArXiv CS.AI 50%

arXiv:2606.30650v1 Announce Type: cross Abstract: Educational support services often face a qualified-capacity problem: staff time is scarce, qualifications decay, new support needs can appear before anyone is prepared for them, and training consumes the same hours needed by...

https://arxiv.org/abs/2606.30650

6 **CONF**

Comparative Analysis of Machine Learning based Intrusion Detection in Realistic IoT Networks

ArXiv CS.AI 50%

arXiv:2606.31594v1 Announce Type: cross Abstract: The Internet of Things (IoT) is rapidly growing and expanding into various sectors, such as healthcare, transportation, smart homes, and more. Despite the benefits of using IoT devices, they present several challenges. Given the...

https://arxiv.org/abs/2606.31594

7 **CONF**

AI Transparency: Governance Compliance or Stakeholder Requirements?

ArXiv CS.AI 50%

arXiv:2606.30652v1 Announce Type: cross Abstract: Transparency is increasingly mandated for public-sector AI systems, with organisations required to publish statements describing their AI use and oversight arrangements. However, the existence of such artefacts is often treated...

https://arxiv.org/abs/2606.30652

8 **CONF**

The Consistency Dilemma in LLMs: Generator-Evaluator Agreement and Vulnerability to Mistakes

ArXiv CS.AI 50%

arXiv:2606.30653v1 Announce Type: cross Abstract: Large language models are increasingly deployed in agentic pipelines that depend on the model evaluating its own outputs without external verification. The reliability of these pipelines depends on an implicit assumption: that...

https://arxiv.org/abs/2606.30653

9 CONF

Toward AI-Resilient Assessment in Computer Science Courses in an AI-Native World

ArXiv CS.AI 50%

arXiv:2606.30655v1 Announce Type: cross Abstract: AI-native course assessments in senior computer science courses and related fields should grade students by \emph{AI-resilient skill}: the ability to achieve outcomes beyond a strong AI baseline. Such assessments should allow...

https://arxiv.org/abs/2606.30655

0 CONF

Mapping the Artificial Intelligence Divide in Africa: Infrastructure, Accessibility and Capacity

ArXiv CS.AI 50%

arXiv:2606.30656v1 Announce Type: cross Abstract: Artificial Intelligence (AI) has the potential to be transformative for development, but Africa is currently facing a fragmented and challenging "AI divide". This paper provides an empirical analysis of the current state of the...

https://arxiv.org/abs/2606.30656

1 CONF

AI for Quality Assurance in the Operating Room

ArXiv CS.AI 50%

arXiv:2606.30657v1 Announce Type: cross Abstract: Surgical outcomes depend not only on patient factors and postoperative care but are also strongly influenced by the quality of the operation itself. Yet, for much of mod-ern surgery, intraoperative quality has been assessed...

https://arxiv.org/abs/2606.30657

2 CONF

Agentic AI Enhances Physician Trust in Clinical Decision Making

ArXiv CS.AI 50%

arXiv:2606.30658v1 Announce Type: cross Abstract: Medical AI has shifted from reasoning to agentic AI, a new paradigm that autonomously invokes external tools during reasoning, rendering intermediate reasoning steps and tool outputs transparent to users. Although proven to...

https://arxiv.org/abs/2606.30658

3 CONF

Improving Survey Participation in Low-Literacy Populations Through Value-Sensitive Conversational AI

ArXiv CS.AI 50%

arXiv:2606.30660v1 Announce Type: cross Abstract: Collecting reliable social data from low-literacy populations remains a persistent challenge, particularly when surveys involve sensitive topics and marginalized communities. Traditional paper-based and web-based survey...

https://arxiv.org/abs/2606.30660

4 CONF

ELEVATE: Designing Human-Centered GenAI Virtual Tutors for Scalable and Inclusive Education

ArXiv CS.AI 50%

arXiv:2606.30662v1 Announce Type: cross Abstract: The advent of Generative Artificial Intelligence (GenAI), and in particular Large Language Models (LLMs), is reshaping educational practice, while intensifying ethical debate about its adoption. To date, the dominant paradigm...

https://arxiv.org/abs/2606.30662

5 **CONF**

Emergent Culture in Minimal LLM Systems

ArXiv CS.AI 50%

arXiv:2606.30668v1 Announce Type: cross Abstract: What happens when LLM agents operate with no context outside a turn, minimal prompting, and simple tools? Inspired by swarm engineering, we give collectives of three agents the ability to send messages and manipulate a shared...

https://arxiv.org/abs/2606.30668

6 **CONF**

Local Pheromone Network: Sparse Local Learning with Multi-Scale Synaptic Trails, Consolidation, and Replay

ArXiv CS.AI 50%

arXiv:2606.30669v1 Announce Type: cross Abstract: Backpropagation-trained dense neural networks are powerful function approximators, but they couple learning across many parameters and can overwrite previous associations when tasks conflict. This paper describes Local Pheromone...

https://arxiv.org/abs/2606.30669

7 **CONF**

Listening Between the Lines: Joint Learning of ASR Embeddings and LLM-Augmented Linguistics for Dementia Detection

ArXiv CS.AI 50%

arXiv:2606.30675v1 Announce Type: cross Abstract: Early detection of dementia through speech analysis offers a non-invasive screening alternative, but capturing both acoustic and linguistic biomarkers remains challenging. We propose a multimodal framework leveraging Whisper for...

https://arxiv.org/abs/2606.30675

8 **CONF**

Optimal Self-Consistency for Efficient Reasoning with Large Language Models

ArXiv CS.AI 50%

arXiv:2511.12309v2 Announce Type: replace-cross Abstract: Self-consistency (SC) is a widely used test-time inference technique for improving performance in chain-of-thought reasoning. It consists of generating multiple responses, or ``samples", from a large language model (LLM)...

https://arxiv.org/abs/2511.12309

9 **CONF**

Unsupervised Thermodynamics of Molecular Diffusion Models: Action-Operator Semantics and Auditable Free-Energy Readout

ArXiv CS.AI 50%

arXiv:2606.30687v1 Announce Type: cross Abstract: Diffusion models are increasingly utilized for modeling molecular structures and conformational ensembles, yet the thermodynamic meaning of their learned representations and scores remains elusive. To resolve this ambiguity, we...

https://arxiv.org/abs/2606.30687

0 **CONF**

Citation Discipline in Spec-Driven Development: A Cross-Model Empirical Study of Output Determinism and Automated Hallucination Detection in LLM-Generated Code

ArXiv CS.AI 50%

arXiv:2606.30689v1 Announce Type: cross Abstract: Spec-Driven Development (SDD) frameworks guide Large Language Model (LLM)-powered code generation through formal specifications, yet they differ fundamentally in how they enforce traceability between requirements and generated...

https://arxiv.org/abs/2606.30689

1 **CONF**

DSIP: A Dynamic Coordination Planner for Signal-Free Intersections using Diffusion-Model-Based Multi-Agent Motion Planning

ArXiv CS.AI 50%

arXiv:2606.30694v1 Announce Type: cross Abstract: Traffic signal control at urban intersections inherently introduces stop-and-go behavior, resulting in increased delays and reduced traffic efficiency, especially under high traffic demand. With the emergence of connected and...

https://arxiv.org/abs/2606.30694

2 **CONF**

IterCAD: An Iterative Multimodal Agent for Visually-Grounded CAD Generation and Editing

ArXiv CS.AI 50%

arXiv:2606.13368v2 Announce Type: replace Abstract: Computer-Aided Design is pivotal in modern manufacturing, yet existing automated methods predominantly rely on open-loop, one-shot generation, creating a mismatch with iterative real-world practices. In this paper, we present...

https://arxiv.org/abs/2606.13368

3 **CONF**

Accelerometry-Derived Digital Biomarkers for Cardiometabolic Risk: A Population-Representative Tabular Benchmark with Uncertainty Quantification

ArXiv CS.AI 50%

arXiv:2606.30702v1 Announce Type: cross Abstract: Structured tabular data dominates clinical medicine, yet existing benchmarks fail to reflect real-world properties like complex survey sampling, demographic oversampling, and subgroup fairness. We introduce the NHANES...

https://arxiv.org/abs/2606.30702

4 **CONF**

From Search to Synthesis: Training LLMs as Zero-Shot Workflow Generators

ArXiv CS.AI 50%

arXiv:2606.30704v1 Announce Type: cross Abstract: Large language models (LLMs) excel across a wide range of tasks, yet their instance-specific solutions often lack the structural consistency needed for reliable deployment. Workflows that encode recurring algorithmic patterns at...

https://arxiv.org/abs/2606.30704

5 **CONF**

Why Do Few-Step Text Latents Fail When Image Latents Work? Non-Commitment at Sharp Categorical Readouts

ArXiv CS.AI 50%

arXiv:2606.30705v1 Announce Type: cross Abstract: Deterministic few-step generation succeeds on continuous image latents but collapses to incoherent text on continuous text latents, and we show the cause is geometric rather than a training or scaling deficiency: a smooth,...

https://arxiv.org/abs/2606.30705

6 **CONF**

WorldRoamBench: An Open-World Benchmark for Long-Horizon Stability of Interactive World Models

ArXiv CS.AI 50%

arXiv:2606.31672v1 Announce Type: cross Abstract: Despite rapid progress in interactive world models (IWMs), existing benchmarks evaluate action following only at trajectory level and ignore memory and interaction physics. We introduce WorldRoamBench, an open-world benchmark for...

https://arxiv.org/abs/2606.31672

7 **CONF**

When transformers learn "impossible" languages, what do they learn?

ArXiv CS.AI **50%**

arXiv:2606.30815v1 Announce Type: cross Abstract: Recent work suggests that transformer language models show a bias towards human languages over unnatural ("impossible") languages argued to be unacquirable by humans. However, this literature has largely based these claims on...

https://arxiv.org/abs/2606.30815

8 **CONF**

AI-Generated PowerShell Malware: An Experimental Framework and Dataset

ArXiv CS.AI **50%**

arXiv:2606.30819v1 Announce Type: cross Abstract: Generative AI has emerged as a significant cybersecurity threat, with several recent attack campaigns leveraging LLMs to generate code for malicious purposes via scripting languages such as PowerShell. Consequently, for...

https://arxiv.org/abs/2606.30819

9 **CONF**

Training Therapeutic Judges and Multi-Agent Systems for Human-Aligned Mental Health Support

ArXiv CS.AI **50%**

arXiv:2606.30887v1 Announce Type: cross Abstract: Large language models show promise for mental health support, yet therapeutic quality improves only when evaluation functions as an actionable control signal rather than a passive metric. We introduce a framework that formulates...

https://arxiv.org/abs/2606.30887

0 **CONF**

Curvature-Guided Module Localization for Low-Rank Detoxification of Backdoored Large Language Models

ArXiv CS.AI **50%**

arXiv:2606.30899v1 Announce Type: cross Abstract: Backdoor attacks pose a serious threat to large language models (LLMs) by causing otherwise benign systems to produce attacker-specified malicious behavior when a hidden trigger is present. In this work, we study post hoc...

https://arxiv.org/abs/2606.30899

1 **CONF**

How Human Feedback Shapes AI-generated Community Notes

ArXiv CS.AI **50%**

arXiv:2606.30905v1 Announce Type: cross Abstract: Community Notes, a bridging-based crowd-sourced fact-checking system, has emerged as a new mechanism for moderating misleading information on social media and has been adopted by major platforms including X, Facebook, Instagram,...

https://arxiv.org/abs/2606.30905

2 **CONF**

Triospect: A Three-Dimensional Framework for Robust Statistical AI-Generated Text Detection Against Diverse Attacks

ArXiv CS.AI **50%**

arXiv:2606.31074v1 Announce Type: cross Abstract: Existing AI-generated text detectors are vulnerable to attacks that manipulate textual characteristics. In this study, we propose a novel Triospect Detection Framework by using additional perspectives of content (core ideas) and...

https://arxiv.org/abs/2606.31074

3 CONF

Wait, am I Being Fair? Characterizing Deductive Stereotyping and Mitigating It with Fair-GCG

ArXiv CS.AI 50%

arXiv:2606.30989v1 Announce Type: cross Abstract: Warning: This paper contains several toxic and offensive statements. While reasoning generally improves fairness in recent large language models (LLMs), failures persist. In this work, we identify a failure mode, deductive...

https://arxiv.org/abs/2606.30989

4 CONF

OTCache: Optimal Transport for Geometry-Aware Caching in Diffusion Models

ArXiv CS.AI 50%

arXiv:2606.31026v1 Announce Type: cross Abstract: We propose OTCache, a training-free framework for accelerating diffusion sampling via caching schedule prediction. Existing graph-based caching methods reduce redundant computation by optimizing shortest-path objectives, but rely...

https://arxiv.org/abs/2606.31026

5 CONF

LLM-Driven Personalities for Decision Making in Emergency Simulations

ArXiv CS.AI 50%

arXiv:2606.31038v1 Announce Type: cross Abstract: For virtual humans to appear believable, they must exhibit agency and spatial awareness while interacting with their environment in ways that reflect competence and intelligence. At the core of these capabilities lies effective...

https://arxiv.org/abs/2606.31038

6 CONF

ADAPT: Attention Dynamics Alignment with Preference Tuning for Faithful MLLMs

ArXiv CS.AI 50%

arXiv:2606.31054v1 Announce Type: cross Abstract: Multimodal Large Language Models (MLLMs) are critically hampered by hallucination, generating content inconsistent with the provided image. In this paper, we identify an internal signature of hallucination: progressive...

https://arxiv.org/abs/2606.31054

7 CONF

When Reranking Hurts: Uncertainty-Based Gating for Few-Shot Reranking

ArXiv CS.AI 50%

arXiv:2606.31087v1 Announce Type: cross Abstract: Few-shot selection typically assumes that reranking retrieved examples always improves performance. We challenge this view by identifying that the expensive reranking step can in fact degrade performance. Instead, we propose...

https://arxiv.org/abs/2606.31087

8 CONF

Seeing Through Multiple Views: Parameter-Efficient Fine-Tuning via Selective Neurons for Consistent Radiology Report Generation

ArXiv CS.AI 50%

arXiv:2606.31099v1 Announce Type: cross Abstract: Recent years have seen substantial advances in radiology report generation (RRG), yet existing approaches predominantly adopt direct feature fusion when handling multi-view X-ray images. Such approaches overlook the potential...

https://arxiv.org/abs/2606.31099

9 CONF

PPT-Eval: A Benchmark for Computer-Use Agents on PowerPoint Tasks

ArXiv CS.AI 50%

arXiv:2606.31154v1 Announce Type: cross Abstract: Creating and editing slides is a rich, multimodal activity that is ubiquitous in professional and educational settings, making it an ideal testbed for real-world computer-use agents. Microsoft PowerPoint is among the most widely...

https://arxiv.org/abs/2606.31154

0 CONF

One Retrieval to Cover Them All: Co-occurrence-Aware Knowledge Base Reorganization for Session-Level RAG

ArXiv CS.AI 50%

arXiv:2606.31156v1 Announce Type: cross Abstract: RAG systems retrieve documents optimized for answering one query at a time. Yet enterprise users arrive with sessions, that is, coherent episodes of related questions that span semantically distant parts of the knowledge base. We...

https://arxiv.org/abs/2606.31156

1 CONF

LLM-Powered Interactive Robotic Action Synthesis from Multimodal Speech, Gestures, and Music

ArXiv CS.AI 50%

arXiv:2606.31158v1 Announce Type: cross Abstract: The quest for intuitive and natural human-robot interaction (HRI) remains a significant challenge in robotics. Traditional methods often rely on rigid, pre-programmed commands that limit the robot's expressiveness and...

https://arxiv.org/abs/2606.31158

2 CONF

ComplianceGate: Classifier-Gated Multi-Tier LLM Routing for Inference in Regulated Industries

ArXiv CS.AI 50%

arXiv:2606.31163v1 Announce Type: cross Abstract: Large language models deployed in regulated industries operate under two constraints: compliance enforcement and cost efficiency. Personally identifiable information (PII) in user queries can reach model endpoints before the...

https://arxiv.org/abs/2606.31163

3 CONF

Transformers as Bayesian In-Context Experimenters: Smoothness-Adaptive Efficient ATE Estimation

ArXiv CS.AI 50%

arXiv:2606.31184v1 Announce Type: cross Abstract: Adaptive experiments for average treatment effects (ATE) require randomized allocations balancing valid inference with statistical efficiency. The oracle design is a covariate-dependent Neyman rule governed by unknown...

https://arxiv.org/abs/2606.31184

4 CONF

Can LLMs Imagine Moral Alternatives Beyond Binary Dilemmas?

ArXiv CS.AI 50%

arXiv:2606.31213v1 Announce Type: cross Abstract: As large language models (LLMs) are increasingly deployed as moral advisors and agents, they need to address dilemmas between two competing values. However, existing research on LLMs with moral dilemmas overlooks a central aspect...

https://arxiv.org/abs/2606.31213

5 CONF

Information-Aided DVL Calibration

ArXiv CS.AI 50%

arXiv:2606.31216v1 Announce Type: cross Abstract: The Doppler velocity log (DVL) velocity measurements are critical to the accuracy of autonomous underwater vehicle (AUV) navigation solutions and, consequently, to mission success. To ensure accurate measurements, the DVL is...

https://arxiv.org/abs/2606.31216

6 CONF

Cross-lingual Relation Extraction with Large Language Models: Zero-Shot, Few-Shot, and Fine-Tuned Evaluation on Romanian

ArXiv CS.AI 50%

arXiv:2606.31718v1 Announce Type: cross Abstract: Relation extraction (RE) for low-resource languages is typically constrained by the lack of annotated corpora. We investigate the feasibility of cross-lingual RE for Romanian by combining automatic dataset translation with large...

https://arxiv.org/abs/2606.31718

7 CONF

SwiftAudio: Data-Efficient Caption-Only Distillation for One-Step Text-to-Audio Diffusion-based Generation

ArXiv CS.AI 50%

arXiv:2606.31259v1 Announce Type: cross Abstract: Diffusion-based text-to-audio (TTA) models achieve impressive synthesis quality but suffer from high inference latency due to iterative multi-step denoising. Existing one-step approaches alleviate this issue but still rely on...

https://arxiv.org/abs/2606.31259

8 CONF

TDGT: A Tabular Data Generation Toolkit supporting adaptive GPU-accelerated Bayesian mixture models, diffusion-based models, and latent-space generative modeling

ArXiv CS.AI 50%

arXiv:2606.31268v1 Announce Type: cross Abstract: The growing demand for privacy-preserving data sharing has positioned synthetic data generation as a critical component of responsible AI workflows. Despite notable advances in generative modeling, existing solutions often lack...

https://arxiv.org/abs/2606.31268

9 CONF

Learning from Failure: Inference-Time Self-Improvement for Computer-Use Agents

ArXiv CS.AI 50%

arXiv:2606.31270v1 Announce Type: cross Abstract: Computer-use agents, which leverage multimodal large language models (MLLMs) to operate computers and complete tasks, have attracted significant attention for their utility and versatility. A major challenge in developing these...

https://arxiv.org/abs/2606.31270

00 CONF

Minimizing Quantized Semantic Age of Information (QSAoI) in Foundation Model-Based Semantic Communications

ArXiv CS.AI 50%

arXiv:2606.31303v1 Announce Type: cross Abstract: The emerging techniques of semantic communications and edge computing in 6G networks necessitate a paradigm shift toward co-designed semantic-aware and adaptive resource allocation for short-packet transmissions. However, there...

https://arxiv.org/abs/2606.31303

01 **CONF**

CSO-LLM: Class Subspace Orthogonalization for Post-Training Backdoor Detection and Trigger Inversion in LLMs

ArXiv CS.AI **50%**

arXiv:2606.31309v1 Announce Type: cross Abstract: While post-training backdoor detection and trigger inversion schemes have been developed for AIs used e.g. for images, there is a paucity of such methods for LLMs. First, the LLM input space is discrete, with up to 150,000^k...

https://arxiv.org/abs/2606.31309

02 **CONF**

From Idea to Prototype in an Afternoon: Scaffolded, AI-Assisted Rapid VA Prototyping

ArXiv CS.AI **50%**

arXiv:2606.31311v1 Announce Type: cross Abstract: Testing a new visual-analytics idea usually takes months: one needs to find a realistic data set, clean it, and implement an interactive prototype. We describe a case where a workflow language and an AI assistant reduced this...

https://arxiv.org/abs/2606.31311

03 **CONF**

PGUDA: Pressure-Guided Unsupervised Domain Adaptation with Cross-Modal Knowledge Distillation for sEMG-Based Gesture Recognition

ArXiv CS.AI **50%**

arXiv:2606.31349v1 Announce Type: cross Abstract: Surface electromyography (sEMG)-based gesture recognition has emerged as a promising technology for natural human-computer interaction. However, its practical deployment remains challenging due to severe performance degradation...

https://arxiv.org/abs/2606.31349

04 **CONF**

Calibrating the Evaluator: Does Probability Calibration Mitigate Preference Coupling in LLM Agent Feedback Loops?

ArXiv CS.AI **50%**

arXiv:2606.31371v1 Announce Type: cross Abstract: When large language model (LLM) agents adapt their behavior through evaluator feedback, systematic evaluator biases propagate into the agent's learned strategy distribution - a phenomenon termed evaluator preference coupling....

https://arxiv.org/abs/2606.31371

05 **CONF**

Compositional Concept-Based Neuron-Level Interpretability for Deep Reinforcement Learning

ArXiv CS.AI **50%**

arXiv:2502.00684v2 Announce Type: replace-cross Abstract: Deep reinforcement learning (DRL) has successfully addressed many complex control problems. However, the neural networks representing policies or values remain opaque, undermining trust in high-stakes applications. While...

https://arxiv.org/abs/2502.00684

06 **CONF**

Position: Collaborative Agentic AI Needs Interoperability Across Ecosystems

ArXiv CS.AI **50%**

arXiv:2505.21550v2 Announce Type: replace-cross Abstract: Collaborative agentic AI is projected to transform entire industries by enabling AI-powered agents to autonomously perceive, plan, and act within digital environments. Yet, current solutions in this field are all built in...

https://arxiv.org/abs/2505.21550

07 **CONF**

Mixture-of-Control: State-Aware Fine-Tuning for Transformer-based Models

ArXiv CS.AI 50%

arXiv:2606.31397v1 Announce Type: cross Abstract: State-based fine-tuning has emerged as a compelling alternative to weight-based adaptation for transformers, updating lightweight controls into states rather than model weights, offering substantial memory savings while retaining...

https://arxiv.org/abs/2606.31397

08 **CONF**

Seeing Is Not Sharing: Some Vision-Language Models Overestimate Common Ground in Asymmetric Dialogue

ArXiv CS.AI 50%

arXiv:2606.31719v1 Announce Type: cross Abstract: In collaborative dialogue, shared perception does not guarantee shared interpretation. Mutual understanding must be established through interaction. We investigate whether vision-language models (VLMs) can distinguish what could...

https://arxiv.org/abs/2606.31719

09 **CONF**

Von Mises Based Uncertainty Quantification for Closely Spaced Automotive Radar Targets

ArXiv CS.AI 50%

arXiv:2606.31473v1 Announce Type: cross Abstract: This work investigates uncertainty-aware deep learning approaches for direction of arrival (DOA) estimation in automotive radar, focusing on probabilistic modeling and downstream integration. A circular-statistics-based von Mises...

https://arxiv.org/abs/2606.31473

10 **CONF**

FinPersona-Bench: A Benchmark for Longitudinal Psychometric Stability of Autonomous Financial Agents

ArXiv CS.AI 50%

arXiv:2606.31522v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly deployed as autonomous financial agents initialized with explicit behavioral mandates such as "preserve capital" or "avoid speculative bets" that are meant to govern every decision...

https://arxiv.org/abs/2606.31522

11 **CONF**

On the Convergence of Self-Improving Online LLM Alignment

ArXiv CS.AI 50%

arXiv:2606.31524v1 Announce Type: cross Abstract: The Self-Improving Alignment (SAIL) algorithm addresses distribution shift by reducing a bilevel formulation of the problem to an efficient, single-level method. Empirically, SAIL has demonstrated strong performance on this task....

https://arxiv.org/abs/2606.31524

12 **CONF**

FLARE-AI: Flaw Reporting for AI

ArXiv CS.AI 50%

arXiv:2606.31567v1 Announce Type: cross Abstract: Flaw reporting for deployed AI systems is fundamental to identifying system failures and improving AI safety. Yet the AI reporting ecosystem is fragmented: researchers who identify flaws often do not know what or where to report,...

https://arxiv.org/abs/2606.31567

13 CONF

Temperature Field Reconstruction of Tungsten Monoblock Divertor on EAST using Physics-aware Neural Operator Transformer

ArXiv CS.AI 50%

arXiv:2606.31574v1 Announce Type: cross Abstract: Accurate modeling of the divertor temperature field is essential for preventing material melting and damage and for extending the service life of fusion devices. However, conventional numerical methods, such as the Finite Element...

https://arxiv.org/abs/2606.31574

14 CONF

ZEBRA: Zero-Shot Entropy-Regularized Prompt Learning for Base-to-Novel Generalization in Audio-Language Models

ArXiv CS.AI 50%

arXiv:2606.31587v1 Announce Type: cross Abstract: Audio-Language Models (ALMs) achieve strong zero-shot performance by aligning audio with textual class descriptions. Although prompt learning improves accuracy on base classes through few-shot supervised adaptation, we observe a...

https://arxiv.org/abs/2606.31587

15 CONF

Evil Spectra: How Optimisers can Amplify or Suppress Emergent Misalignment

ArXiv CS.AI 50%

arXiv:2606.31591v1 Announce Type: cross Abstract: Emergent misalignment (EM) is a recently discovered phenomenon in LLMs where fine-tuning on a narrow misaligned task, such as writing insecure code, leads to broadly misaligned behaviour on unrelated prompts. Previous work has...

https://arxiv.org/abs/2606.31591

16 CONF

Preserve the Hard, Regenerate the Rest: Uncertainty-Guided Synthetic Training Data Augmentation with Diffusion Models

ArXiv CS.AI 50%

arXiv:2606.31603v1 Announce Type: cross Abstract: Semantic segmentation models struggle with data sparsity and rare or visually diverse regions, e.g., dense regions or small objects in aerial or autonomous mobility data. While synthetic augmentation is an appealing solution,...

https://arxiv.org/abs/2606.31603

17 CONF

A Tutorial on Autonomous Fault-Tolerant Control Using Knowledge-Grounded LLM Agents

ArXiv CS.AI 50%

arXiv:2606.31635v1 Announce Type: cross Abstract: Fault recovery in process plants still relies heavily on plant operators, especially when faults fall outside predefined supervisory logic. Operators interpret alarms, procedures, P\&IDs, interlocks, and process trends, then...

https://arxiv.org/abs/2606.31635

18 CONF

A Lifecycle and Application-Stack Survey of Large Language Model Vulnerabilities: Attacks, Risks, Defenses, and Open Problems

ArXiv CS.AI 50%

arXiv:2606.31639v1 Announce Type: cross Abstract: Large language models are no longer only text generators. They are increasingly embedded in retrieval pipelines, enterprise assistants, coding environments, robotic systems, security-operation workflows, and autonomous agents...

https://arxiv.org/abs/2606.31639

19 **CONF**

STEB: Style Text Embedding Benchmark

ArXiv CS.AI 50%

arXiv:2606.31741v1 Announce Type: cross Abstract: While semantic embeddings are rigorously evaluated on the Massive Text Embedding Benchmark, the evaluation of style embeddings remains fragmented, with each work relying on their own set of tasks and datasets. To bridge this gap,...

https://arxiv.org/abs/2606.31741

20 **CONF**

JL1-CC&QA: Extending the JL1-CD Benchmark with Change Captioning and Question Answering

ArXiv CS.AI 50%

arXiv:2606.31745v1 Announce Type: cross Abstract: Remote sensing change detection (CD) traditionally focuses on pixel-level binary segmentation, which identifies where changes occur but neither what nor why. To bridge this semantic gap, we introduce JL1-CC&QA, a multi-task...

https://arxiv.org/abs/2606.31745

21 **CONF**

A Technical Typology of AI Systems in Public Administration

ArXiv CS.AI 50%

arXiv:2606.31755v1 Announce Type: cross Abstract: Research on artificial intelligence (AI) in the public sector often treats "AI" as a single category, neglecting technical distinctions between different AI systems. But these distinctions affect how different systems impact core...

https://arxiv.org/abs/2606.31755

22 **CONF**

Attend, Transform, or Silence: Operator-Level Visual Skipping for Efficient Multimodal LLM Inference

ArXiv CS.AI 50%

arXiv:2606.31903v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) increasingly process long visual-token sequences, increasing the overall inference computation. Existing acceleration methods usually remove visual tokens or skip visual-token updates in...

https://arxiv.org/abs/2606.31903

23 **CONF**

MECoBench: A Systematic Study of Multimodal Agent Collaboration in Embodied Environments

ArXiv CS.AI 50%

arXiv:2606.31966v1 Announce Type: cross Abstract: Recent multimodal large language models (MLLMs) have strong potential as embodied agents, but their ability to collaborate in visually grounded environments remains underexplored. To address this gap, we introduce MECoBench, a...

https://arxiv.org/abs/2606.31966

24 **CONF**

Amplifying Membership Signal Through Chained Regeneration

ArXiv CS.AI 50%

arXiv:2606.31991v1 Announce Type: cross Abstract: The tendency of large generative models to memorize training data makes sample verification critical for privacy auditing and copyright enforcement. Current membership (MIA) and dataset inference (DI) attacks often rely on...

https://arxiv.org/abs/2606.31991

25 CONF

FLORA: A deep learning approach to predict forest attributes from heterogeneous LiDAR data

ArXiv CS.AI 50%

arXiv:2606.32023v1 Announce Type: cross Abstract: Forest attributes are essential for national-scale resource monitoring. Airborne LiDAR metrics are among the auxiliary variables most strongly correlated with forest attributes used in National Forest Inventory (NFI) estimates....

https://arxiv.org/abs/2606.32023

26 CONF

When LLMs Read Tables Carelessly: Measuring and Reducing Data Referencing Errors

ArXiv CS.AI 50%

arXiv:2606.32029v1 Announce Type: cross Abstract: While large language models (LLMs) perform well on table tasks, they still make data referencing errors (DREs), i.e., incorrectly citing or omitting table values, despite understanding the table structure. Beyond final-answer...

https://arxiv.org/abs/2606.32029

27 CONF

Reinforcement Learning with Metacognitive Feedback Elicits Faithful Uncertainty Expression in LLMs

ArXiv CS.AI 50%

arXiv:2606.32032v1 Announce Type: cross Abstract: Metacognition is a critical component of intelligence that describes the ability to monitor and regulate one's own cognitive processes. Yet LLMs exhibit systemic deficiencies in key metacognitive faculties: they hallucinate with...

https://arxiv.org/abs/2606.32032

28 CONF

QVal: Cheaply Evaluating Dense Supervision Signals for Long-Horizon LLM Agents

ArXiv CS.AI 50%

arXiv:2606.32034v1 Announce Type: cross Abstract: LLM agents increasingly act over long horizons, where a single trajectory can contain hundreds or thousands of actions. In these settings, outcome-only rewards provide too sparse guidance, failing to inform the model about the...

https://arxiv.org/abs/2606.32034

29 CONF

Introspective Coupling: Self-Explanation Training Tracks Behavioral Change Despite Fixed Supervision

ArXiv CS.AI 50%

arXiv:2606.32038v1 Announce Type: cross Abstract: When does training language models (LMs) to generate explanations of their predictions yield faithful introspection, rather than superficial imitation? We study LMs trained to explain which features of their inputs influenced...

https://arxiv.org/abs/2606.32038

30 CONF

Deductive Logic in Language Models: Horizontal vs Vertical Reasoning

ArXiv CS.AI 50%

arXiv:2510.09340v2 Announce Type: replace Abstract: Recent language models exhibit significant logical reasoning abilities, yet the mechanisms supporting deductive inference remain poorly understood. This paper studies small transformer-based language models trained from scratch...

https://arxiv.org/abs/2510.09340

31 CONF

LLM-Empowered Agentic MAC Protocols: A Dynamic Stackelberg Game Approach

ArXiv CS.AI 50%

arXiv:2510.10895v2 Announce Type: replace Abstract: Medium Access Control (MAC) protocols, essential for wireless networks, are typically manually configured. While deep reinforcement learning (DRL)-based protocols enhance task-specified network performance, they suffer from...

https://arxiv.org/abs/2510.10895

32 CONF

Improving LLM Reasoning with Homophily-aware Structural and Semantic Text-Attributed Graph Compression

ArXiv CS.AI 50%

arXiv:2601.08187v3 Announce Type: replace Abstract: Large language models (LLMs) have demonstrated promising capabilities in Text-Attributed Graph (TAG) understanding. Recent studies typically focus on verbalizing the graph structures via handcrafted prompts, feeding the target...

https://arxiv.org/abs/2601.08187

33 CONF

Diffusion Crossover: Defining Evolutionary Recombination in Diffusion Models via Noise Sequence Interpolation

ArXiv CS.AI 50%

arXiv:2604.14790v2 Announce Type: replace Abstract: Interactive Evolutionary Computation (IEC) provides a powerful framework for optimizing subjective criteria such as human preferences and aesthetics, yet it suffers from a fundamental limitation: in high-dimensional generative...

https://arxiv.org/abs/2604.14790

34 CONF

LiteResearcher: A Scalable Agentic RL Training Framework for Deep Research Agent

ArXiv CS.AI 50%

arXiv:2604.17931v3 Announce Type: replace Abstract: Reinforcement Learning (RL) has emerged as a powerful training paradigm for LLM-based agents. However, scaling agentic RL for deep research remains constrained by two coupled challenges: hand-crafted synthetic data fails to...

https://arxiv.org/abs/2604.17931

35 CONF

Containment Verification: AI Safety Guarantees Independent of Alignment

ArXiv CS.AI 50%

arXiv:2605.09045v2 Announce Type: replace Abstract: Agentic frameworks are the software layer through which AI agents act in the world. Existing safety methods intervene on the model and therefore remain conditional on unverifiable properties of learned behavior. We introduce...

https://arxiv.org/abs/2605.09045

36 CONF

Reasoning4Sciences: Bridging Reasoning Language Models to All Scientific Branches

ArXiv CS.AI 50%

arXiv:2606.01145v3 Announce Type: replace Abstract: While Reasoning Language Models (RLMs) are rapidly emerging as powerful tools for scientific research, their impact is primarily concentrated in "hard science" fields. The slow -- or lack of -- adoption of RLMs in other...

https://arxiv.org/abs/2606.01145

37 CONF

IPO Finance Agent: Benchmark of LLM Financial Analysts Beyond Finance Agent v2, with Automated Rubric Generation, on the SpaceX (SPCX) IPO

ArXiv CS.AI 50%

arXiv:2606.23032v3 Announce Type: replace Abstract: Finance Agent v2 (by Vals AI) has emerged as the reference benchmark for evaluating both Anthropic Claude and OpenAI ChatGPT frontier language models on financial tasks. However, it narrowly deals with periodic reporting from...

https://arxiv.org/abs/2606.23032

38 CONF

Teaching LLMs String Matching, Backtracking, and Error Recovery to Deduce Bases and Truth Tables for the Combinatorially Exploding Bit Manipulation Puzzles

ArXiv CS.AI 50%

arXiv:2606.23672v2 Announce Type: replace Abstract: This paper presents our algorithmic innovations for the NVIDIA Nemotron Model Reasoning Challenge, focusing on Bit Manipulation Puzzles. In this task, the objective is to discover a hidden logical rule transforming input binary...

https://arxiv.org/abs/2606.23672

39 CONF

Perturbation Effects on Robustness and Individual Fairness

ArXiv CS.AI 50%

arXiv:2404.01356v4 Announce Type: replace-cross Abstract: Deep neural networks are vulnerable to adversarial perturbations that can simultaneously degrade prediction robustness and individual fairness across diverse application settings. However, existing evaluation protocols...

https://arxiv.org/abs/2404.01356

40 CONF

Verify when Uncertain: Beyond Self-Consistency in Black Box Hallucination Detection

ArXiv CS.AI 50%

arXiv:2502.15845v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) often hallucinate, limiting their reliability in sensitive applications. In black-box settings, several self-consistency-based techniques have been proposed for hallucination detection. We...

https://arxiv.org/abs/2502.15845

41 CONF

Artificial Intelligence in Sports: Insights from a Quantitative Survey among Sports Students in Germany about their Perceptions, Expectations, and Concerns regarding the Use of AI Tools

ArXiv CS.AI 50%

arXiv:2503.05785v2 Announce Type: replace-cross Abstract: Generative Artificial Intelligence (AI) tools such as ChatGPT, Copilot, or Gemini have a crucial impact on academic research and teaching. Empirical data on how students perceive the increasing influence of AI, which...

https://arxiv.org/abs/2503.05785

42 CONF

TraCeS: Learning Per-Timestep Constraint-Violation Credit from Sparse Trajectory-Level Labels

ArXiv CS.AI 50%

arXiv:2504.12557v3 Announce Type: replace-cross Abstract: Ensuring safe behavior in reinforcement learning (RL) is challenging when safety constraints are implicit and cannot be densely measured. In many settings, supervision is limited to coarse approvals or rejections of whole...

https://arxiv.org/abs/2504.12557

43 CONF

A Reproducible Benchmark of Lightweight CNNs: Accuracy, Efficiency, and the Impact of Pretrained Initialization

ArXiv CS.AI 50%

arXiv:2505.03303v3 Announce Type: replace-cross Abstract: Lightweight convolutional neural networks are often compared using results obtained with different training recipes, input settings, and pretrained checkpoints. Such differences make architecture rankings difficult to...

https://arxiv.org/abs/2505.03303

44 CONF

From Multimodal Perception to Strategic Reasoning: A Survey on AI-Generated Game Commentary

ArXiv CS.AI 50%

arXiv:2506.17294v3 Announce Type: replace-cross Abstract: The advent of artificial intelligence has propelled AI-Generated Game Commentary (AI-GGC) into a rapidly expanding research area, offering advantages such as scalable availability and personalized narration. However,...

https://arxiv.org/abs/2506.17294

45 CONF

Physics-Constrained Fine-Tuning of Flow-Matching Models for Generation and Inverse Problems

ArXiv CS.AI 50%

arXiv:2508.09156v3 Announce Type: replace-cross Abstract: We present a framework for fine-tuning flow-matching generative models to enforce physical constraints and solve inverse problems in scientific systems. Starting from a model trained on low-fidelity or observational data,...

https://arxiv.org/abs/2508.09156

46 CONF

Dataset Construction for Training LLM to Learn Analog Circuit Knowledge

ArXiv CS.AI 50%

arXiv:2508.10409v3 Announce Type: replace-cross Abstract: This paper constructs a textual dataset for training large language models (LLMs) to learn analog circuit knowledge and customizes LLM training techniques. For dataset construction, high-quality textbooks are collected...

https://arxiv.org/abs/2508.10409

47 CONF

CharDiff-LP: A Diffusion Model with Character-Level Guidance for License Plate Image Restoration

ArXiv CS.AI 50%

arXiv:2510.17330v3 Announce Type: replace-cross Abstract: License plate image restoration is important not only as a preprocessing step for license plate recognition but also for enhancing evidential value, improving visual clarity, and enabling broader reuse of license plate...

https://arxiv.org/abs/2510.17330

48 CONF

Rethinking Garment Conditioning in Diffusion-based Virtual Try-On: Decouple, Don't Denoise

ArXiv CS.AI 50%

arXiv:2511.18775v2 Announce Type: replace-cross Abstract: Virtual Try-On (VTON) synthesizes realistic images of a person wearing a target garment, with broad applications in e-commerce and fashion. Diffusion-based dual-UNet methods achieve strong results but double the...

https://arxiv.org/abs/2511.18775

49 **CONF**

InfiniteWeb: Scalable Web Environment Synthesis for GUI Agent Training

ArXiv CS.AI 50%

arXiv:2601.04126v3 Announce Type: replace-cross Abstract: GUI agents that interact with graphical interfaces on behalf of users represent a promising direction for practical AI assistants. However, training such agents is hindered by the scarcity of suitable environments. We...

https://arxiv.org/abs/2601.04126

50 **CONF**

From Similarity to Vulnerability: Key Collision Attack on LLM Semantic Caching

ArXiv CS.AI 50%

arXiv:2601.23088v2 Announce Type: replace-cross Abstract: Semantic caching has emerged as a pivotal technique for scaling LLM applications, widely adopted by major providers including AWS and Microsoft. By utilizing semantic embedding vectors as cache keys, this mechanism...

https://arxiv.org/abs/2601.23088

51 **CONF**

Reward Redistribution for CVaR MDPs using a Bellman Operator on L-infinity

ArXiv CS.AI 50%

arXiv:2602.03778v2 Announce Type: replace-cross Abstract: Tail-end risk measures such as static conditional value-at-risk (CVaR) are used in safety-critical applications to prevent rare, yet catastrophic events. Unlike risk-neutral objectives, the static CVaR of the return...

https://arxiv.org/abs/2602.03778

52 **CONF**

A swap-adversarial framework for improving domain generalization in electrocorticography-based Parkinson's disease classification

ArXiv CS.AI 50%

arXiv:2602.10528v2 Announce Type: replace-cross Abstract: We propose a novel swap-adversarial framework that mitigates high inter-subject variability and the high-dimensional low-sample-size problem in electrocorticography (ECoG) data. It achieves robust domain generalization...

https://arxiv.org/abs/2602.10528

53 **CONF**

RCTs for Frontier AI Governance: Methodological Challenges and Solutions for Human Uplift Studies

ArXiv CS.AI 50%

arXiv:2603.11001v3 Announce Type: replace-cross Abstract: Human uplift studies, or studies that measure the effects of AI access on human performance via randomized controlled trials (RCT) or similar methodologies, increasingly inform frontier AI governance and deployment...

https://arxiv.org/abs/2603.11001

54 **CONF**

Finite Difference Flow Optimization for RL Post-Training of Text-to-Image Models

ArXiv CS.AI 50%

arXiv:2603.12893v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) has become a standard technique for post-training diffusion-based image synthesis models, as it enables learning from reward signals to explicitly improve desirable aspects such as image...

https://arxiv.org/abs/2603.12893

55 **CONF**

An Efficient Heterogeneous Co-Design for Fine-Tuning on a Single GPU

ArXiv CS.AI **50%**

arXiv:2603.16428v2 Announce Type: replace-cross Abstract: Fine-tuning Large Language Models (LLMs) has become essential for domain adaptation, but its memory-intensive property exceeds the capabilities of most GPUs. To address this challenge and democratize LLM fine-tuning, we...

https://arxiv.org/abs/2603.16428

56 **CONF**

Structured Progressive Knowledge Activation for LLM-Driven Neural Architecture Search

ArXiv CS.AI **50%**

arXiv:2605.04057v3 Announce Type: replace-cross Abstract: This paper focuses on a key challenge in Neural Architecture Search (NAS): integrating established architectural knowledge while exploring new designs under expensive evaluations. Large language models (LLMs) are a...

https://arxiv.org/abs/2605.04057

57 **CONF**

TraceLab: Characterizing Coding Agent Workloads for LLM Serving

ArXiv CS.AI **50%**

arXiv:2606.30560v2 Announce Type: replace-cross Abstract: Coding agents are rapidly becoming a major application of agentic LLMs, but serving them efficiently remains challenging. Progress on this challenge requires understanding real workload patterns, yet the data needed for...

https://arxiv.org/abs/2606.30560

58 **CONF**

Quantitative Movement Testing: Measuring Chronic Pain Patient Movements from a Single Smartphone Video

ArXiv CS.AI **50%**

arXiv:2606.02301v2 Announce Type: replace-cross Abstract: Chronic pain diminishes quality of life by decreasing functional ability, yet objectively measuring this functional impact remains challenging in real-world settings. While optical motion capture provides high precision...

https://arxiv.org/abs/2606.02301

59 **CONF**

The Geometry of Refusal: Linear Instability in Safety-Aligned LLMs

ArXiv CS.AI **50%**

arXiv:2606.22686v2 Announce Type: replace-cross Abstract: Modern Large Language Models (LLMs) rely on extensive safety alignment, yet the mechanistic basis of refusal remains opaque. In this work, we investigate whether safety compliance is a deep semantic decision or a...

https://arxiv.org/abs/2606.22686

60 **CONF**

Brevity is the Soul of Inference Efficiency: Inducing Concision in VLMs via Data Curation

ArXiv CS.AI **50%**

arXiv:2606.25432v2 Announce Type: replace-cross Abstract: Inference efficiency is typically pursued by shrinking the model: distillation, pruning, quantization, and sparse routing each lower per-token cost while treating token count as fixed. But output length has been...

https://arxiv.org/abs/2606.25432

61 CONF

LWDrive: Layer-Wise World-Model-Guided Vision-Language Model Planning for Autonomous Driving

ArXiv CS.AI 50%

arXiv:2606.29879v2 Announce Type: replace-cross Abstract: Vision-Language Models (VLMs) provide powerful semantic understanding and commonsense reasoning for End-to-End Autonomous Driving (E2E-AD) planning. However, trajectories directly generated by VLMs often encode only...

https://arxiv.org/abs/2606.29879